

- 4 (a) $v^2 = 2gh$
 $v^2 = 2 \times 9.8 \times 1.6$ C1
 $v = 5.6 \text{ m s}^{-1}$ A1 [2]
- (b) (i) working leading to idea that $h = 0.90 \times 1.6$ C1
 $h = 1.44 \text{ m}$ A1
(ii) $mgh = \frac{1}{2}mv^2$
 $v^2 = 2 \times 9.8 \times 1.44$ C1
 $v = 5.3 \text{ m s}^{-1}$ A1 [4]
- (c) $\Delta p = m(v - u)$ OR $p = mv$ C1
 $m = 0.073 \text{ kg}$
 $\Delta p = 0.073 \times (5.6 + 5.3)$ C1
 $= 0.80 \text{ N s}$ A1 [3]
- (d) steel plate (and Earth) B1
must gain momentum of 0.80 N s M1
in downward direction A1 [3]
(idea of Earth/plate and ball as the system scores 1/3)